Review 2 – Solutions

① $X \sim N(\theta, 4)$

prior $\theta \sim N(2,2)$

$x = 2$

posterior? MAP?

Bayes rule (CTS) case:

$$f_{\theta|X}(\theta|x) = \frac{f(x|\theta)\cdot f(\theta)}{\int f(x|\theta')\cdot f(\theta')\,d\theta'}$$

θ is mean of normal distribution $\Rightarrow$ its domain is $[-\infty, \infty]$

$$f(\theta|x) = \frac{\frac{1}{\sqrt{2\pi 2}}\exp(-(\theta-2)^2/4)\cdot\frac{1}{\sqrt{2\pi 4}}\cdot\exp(-(x-\theta)^2/8)}{\int \ldots d\theta'} =$$

$$= \text{const}\cdot \exp(-(\theta-2)^2/4)\cdot\exp(-(2-\theta)^2/8) =$$

$$= \text{const}\cdot\exp\left(-\frac{(\theta-2)^2(2+1)}{8}\right) = \text{const}\cdot\exp\left(-\frac{(\theta-2)^2}{2\cdot 4/3}\right)$$

Since $f(\theta|x)$ is a density function, it must be normal (which is also $\text{const}\cdot\exp\left(-\frac{(\theta-\mu)^2}{2\sigma^2}\right)$)

with $\mu = 2$, $\sigma^2 = 4/3$

So, posterior distribution is $\frac{1}{\sqrt{2\pi 4/3}}\cdot\exp\left(-\frac{(\theta-2)^2}{8/3}\right)$

$\boxed{\theta \sim N(2, \tfrac{4}{3})}$ ⊛ same mean as prior, but smaller var – we become more confident!

! In Bayesian problems, make sure twice before computing integral in the denominator that you actually need to compute it! It might take lots of time and be unnecessary!

MAP estimate $\hat{\theta} = \arg\max_{\theta} f(\theta | x) =$

$= \arg\max_{\theta} \left(c \cdot \exp\left(- \frac{(\theta - 2)^2}{8/3} \right) \right) = 2$

at $\theta = 2$ $\exp\left(- \frac{(\theta - 2)^2}{8/3} \right) = 1$,
otherwise it's < 1

$\boxed{\hat{\theta} = 2}$

gaussian density achieves its max when $\theta = \mu$